Optimization for Transportation Planning and Periodic Task Scheduling

Rail Freight Crew Management under Train Delays, Line Planning, and Balanced Assignments

Supervised by Frédéric Meunier (CERMICS) and Juliette Pouzet (SNCF)

Héloïse Gachet

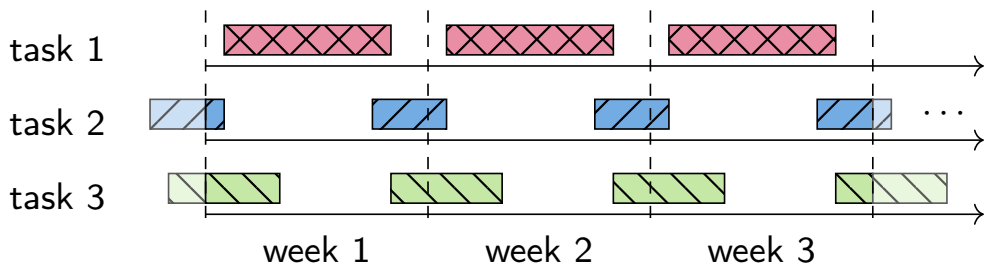
CERMICS, ENPC, IP Paris | November 7, 2026

CERMICS
École nationale des ponts et chaussées

SNCF

Hexafret

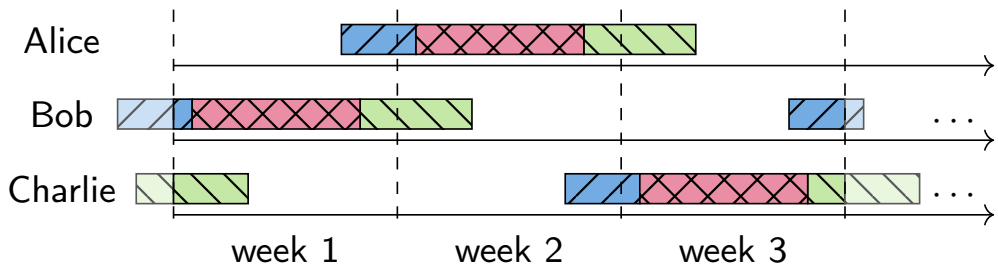
Periodic tasks and assignments



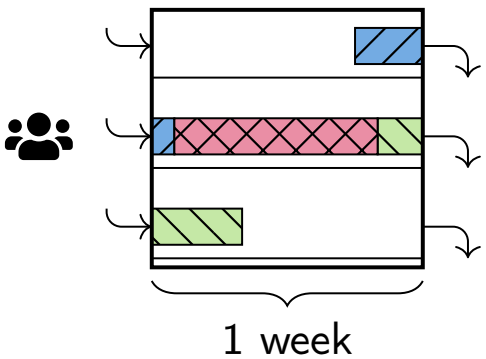
Resources:

Alice
Bob
Charlie
?
?

Aim: Assign tasks to resources for every week



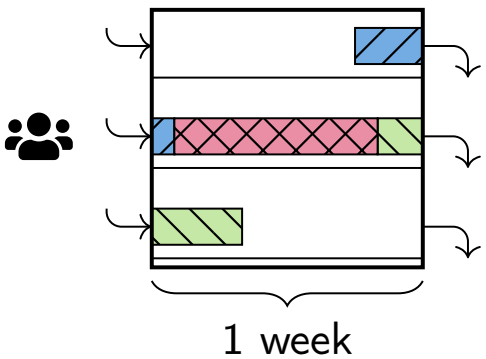
Crew management and balanced assignments



Cyclic rosters:

- ensure balancedness between workers;
- are easy to read and memorize;
- keep workers' proficiency level on every task.

Crew management and balanced assignments

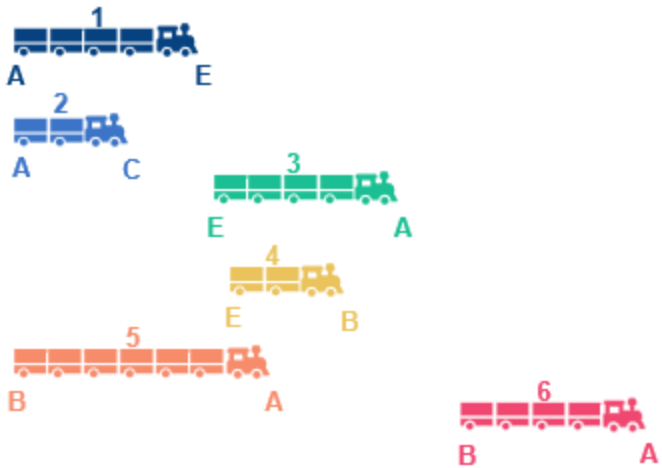


Cyclic rosters:

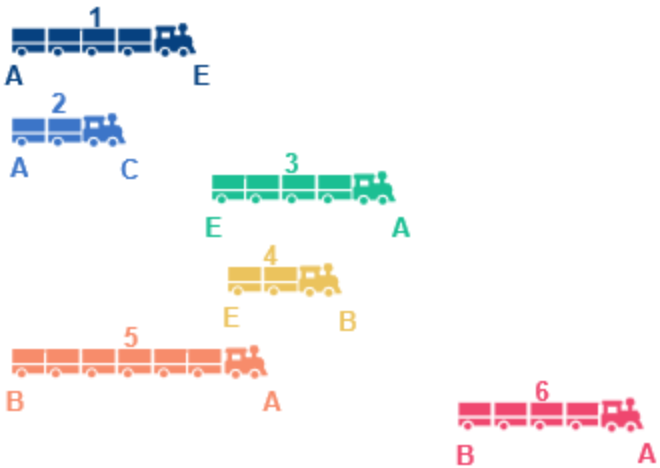
- ensure balancedness between workers;
- are easy to read and memorize;
- keep workers' proficiency level on every task.

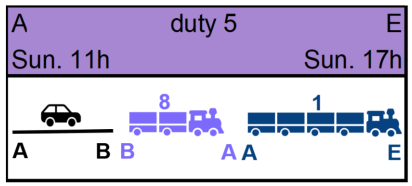
	Crew management	Balanced assignments
Tasks	Trains	Intervals
Structure	Cyclic	Balanced
Number of workers	Not fixed	Fixed
Constraints	Rich	Simple
Goal	Method for efficient solutions	Theorems
Evaluation	Numerical experiments	Proofs

Trains, duties, blocks, and rosters



Trains, duties, blocks, and rosters





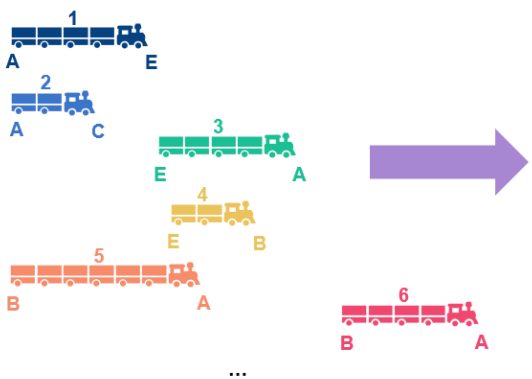
Mon.	Tue.	Wed.
block 1		
duty 7	duty 1	on-call
10h-17h A - A	12h-21h A - A	

Mon.	Tue.	Wed.	Thu.	Fri.	Sat.	Sun.
block 1			R	R	R	block 2
duty 7	duty 1	on-call				duty 5
10h-17h A - A	12h-21h A - A					11h-17h A - E
block 2					R	R
duty 6	duty 2	duty 3	duty 4	on-call		
5h-16h E - A	12h-19h A - B	5h-11h B - A	5h-12h A - A			
R	block 3			R	R	R
	duty 9	duty 10	on-call			
	11h-17h A - E	5h-16h E - A				

Crew management: problem statement

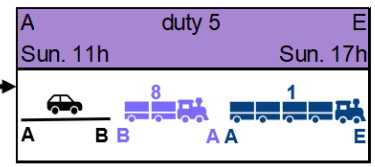
Input. Trains to be driven every week, teams

Output. Covering of trains by rosters with minimum number of working days, one roster per team



Team 1:

Mon.	Tue.	Wed.	Thu.	Fri.	Sat.	Sun.
block 1			R	R	R	block 2
duty 7	duty 1	on-call				duty 5
10h-17h	12h-21h					11h-17h
A - A	A - A					A - E
block 2					R	R
duty 6	duty 2	duty 3	duty 4	on-call		
5h-16h	12h-19h	5h-11h	5h-12h			
E - A	A - B	B - A	A - A			
R	block 3			R	R	R
	duty 9	duty 10	on-call			
	11h-17h	5h-16h				
	A - E	E - A				



Team 2:

Mon.	Tue.	Wed.	Thu.	Fri.	Sat.	Sun.
block 3				R	R	R
duty 11	duty 12	duty 13	on-call			
12h-20h	11h-17h	5h-12h				
B - B	B - B	E - B				

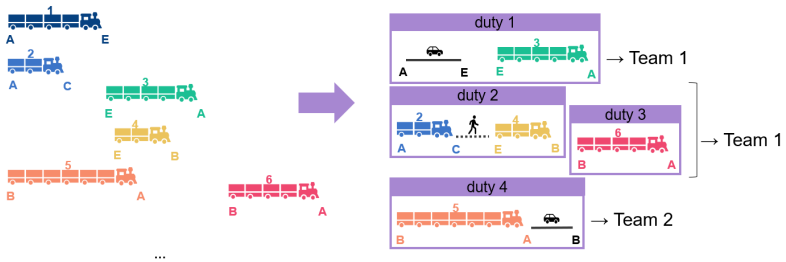
Number of working days = 13

Current heuristic at SNCF: two-stage approach

1. Crew Scheduling

Input. Trains to be driven every week, teams

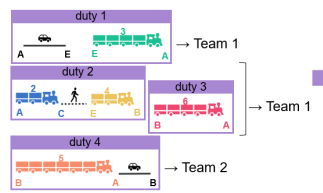
Output. Covering of trains by daily duties with minimum surrogate cost, each duty assigned to a team



2. Crew Rostering

Input. Duties, each assigned to a team

Output. Covering of trains by rosters with minimum number of working days, one roster per team



Team 1:

Mon.	Tue.	Wed.	Thu.	Fri.	Sat.	Sun.
block 1			R	R	block 2	
duty 7	duty 2	duty 3			on-call	duty 5
10h-17h A - A	12h-19h A - B	11h-18h B - A				5h-11h A - E
block 2						
duty 6	duty 1	on-call	duty 4	R	R	R
11h-18h E - A	12h-21h A - A		5h-12h A - A			
block 3						
on-call	duty 9	duty 10	on-call	R	R	R
	11h-17h A - E	5h-16h E - A				

Team 2:

Mon.	Tue.	Wed.	Thu.	Fri.	Sat.	Sun.
block 3						
duty 10	duty 8	duty 9	on-call	R	R	R
12h-20h B - B	11h-17h B - B	5h-12h E - B				

Number of working days = 14

→ Leads to sub-optimal solutions

→ Hard to account for uncertainties

From blocks to rosters

Team 1:

Mon.	Tue.	Wed.
block 1		
duty 7	duty 1	on-call
10h-17h A - A	12h-21h A - A	

Sun.	Mon.	Tue.	Wed.	Thu.	Fri.
block 2					
duty 5	duty 6	duty 2	duty 3	duty 4	on-call
11h-17h A - E	5h-16h E - A	12h-19h A - B	5h-11h B - A	5h-12h A - A	

Tue.	Wed.	Thu.
block 3		
duty 9	duty 10	on-call
11h-17h A - E	5h-16h E - A	

Number of working days = 10

Lemma 3.4.2 (p.45) always possible

Team 1:

Mon.	Tue.	Wed.	Thu.	Fri.	Sat.	Sun.
block 1						block 2
duty 7	duty 1	on-call	R	R	R	duty 5
10h-17h A - A	12h-21h A - A					11h-17h A - E
block 2						
duty 6	duty 2	duty 3	duty 4	on-call	R	R
5h-16h E - A	12h-19h A - B	5h-11h B - A	5h-12h A - A			
R	block 3					
	duty 9	duty 10	on-call	R	R	R
	11h-17h A - E	5h-16h E - A				

Number of working days = 10

Crew management problem:

Input. Trains to be driven every week, teams

Output. Covering of trains by blocks with minimum number of working days, each block assigned to a team

Modeling the crew management problem

- w_b: number of working days in block b
- I: set of teams
- B_i: set of all feasible blocks for team i
- T: collection of trains

$$\begin{aligned} \text{minimize} \quad & \sum_{i \in I} \sum_{b \in B_i} w_b x_{b,i} \\ \text{s.t.} \quad & \sum_{i \in I} \sum_{b \in B_i : t \in b} x_{b,i} \geqslant 1 \qquad \forall t \in T \\ & x_{b,i} \in \{0,1\} \qquad \forall i \in I, \forall b \in B_i \end{aligned}$$

- $x_{b,i} \in \{0,1\}$: block $b \in B_i$ is selected for team $i \in I$

Modeling the crew management problem

- w_b: number of working days in block b
- I: set of teams
- B_i: set of all feasible blocks for team i
- T: collection of trains

$$\begin{aligned} \text{minimize} \quad & \sum_{i \in I} \sum_{b \in B_i} w_b x_{b,i} \\ \text{s.t.} \quad & \sum_{i \in I} \sum_{b \in B_i : t \in b} x_{b,i} \geqslant 1 \qquad \forall t \in T \\ & x_{b,i} \in \{0,1\} \qquad \forall i \in I, \forall b \in B_i \end{aligned}$$

- $x_{b,i} \in \{0,1\}$: block $b \in B_i$ is selected for team $i \in I$

$$\sum_{i \in I} |B_i| \approx 10^{25}$$

Column generation

Column generation: standard exact method for LP with a large number of variables

- Model with an inherently large number of variables
- Reformulation of a compact model (sometimes) yielding stronger bounds

Column generation

Column generation: standard exact method for LP with a large number of variables

- Model with an inherently large number of variables
- Reformulation of a compact model (sometimes) yielding stronger bounds

To keep it short: iteratively generate "interesting" variables with proof of optimality (without all 10^{25} variables)

- "interesting" $\rightarrow$ solving optimization problem (pricing subproblem)

Column generation

Column generation: standard exact method for LP with a large number of variables

- Model with an inherently large number of variables
- Reformulation of a compact model (sometimes) yielding stronger bounds

To keep it short: iteratively generate "interesting" variables with proof of optimality (without all 10^{25} variables)

- "interesting" $\rightarrow$ solving optimization problem (pricing subproblem)

For ILP: column generation-based matheuristic (or exact methods: branch-and-price)

Column generation

Column generation: standard exact method for LP with a large number of variables

- Model with an inherently large number of variables
- Reformulation of a compact model (sometimes) yielding stronger bounds

To keep it short: iteratively generate "interesting" variables with proof of optimality (without all 10^{25} variables)

- "interesting" $\rightarrow$ solving optimization problem (pricing subproblem)

For ILP: column generation-based matheuristic (or exact methods: branch-and-price)

For our problem:

Pricing subproblem = Shortest path with constraints

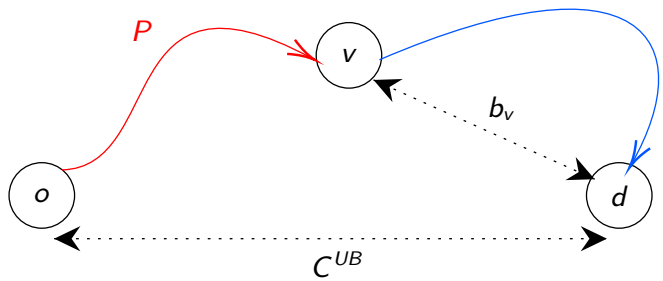
Directed graph:
- vertices = trains
- arc = trains driven successively within a same block
- feasible path = block

Resource Constrained Shortest Path Problem (RCSPP), Section 4.5.2

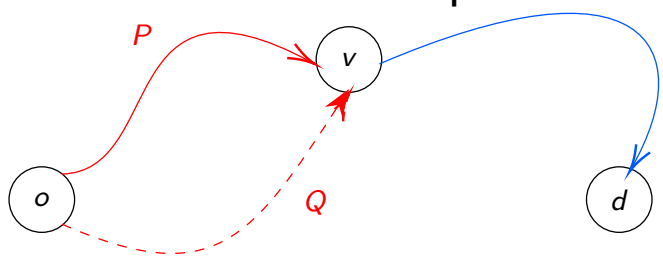
Resource constrained shortest path problem: algorithm

Resolution: path enumeration algorithm using

- Bounds:[1] under-estimate of the resource to reach d



- Dominance:[1] comparison of resources for paths with same last vertex



- Other acceleration techniques: Section 4.5.3

[1] synthetized by Parmentier, A.: **Algorithms for Non-Linear and Stochastic Resource Constrained Shortest Paths**, ***Mathematical Methods of Operations Research***. **(2017)**

Numerical experiments

- Run with:
  - Intel(R) Xeon(R) CPU E5-2690 v2 @ 3.00GHz processor
  - programming language Java 18
  - solver CPLEX 22.11
- Integer linear program time limit 1 h

	Two-stage approach		Integrated approach			
	Time	Best found solution	Time	LB		Best found solution
116 trains	1 min	50	1 h 8 min	40.3	(-18%)	41
280 trains	3 min	228	1 h 9 min	205.3	(-9%)	207
926 trains	26 min	832	6 h 3 min	711.9	(-13%)	724
1,815 trains	64 min	1,633	10 h 2 min	1,395.0	(-10%)	1,477

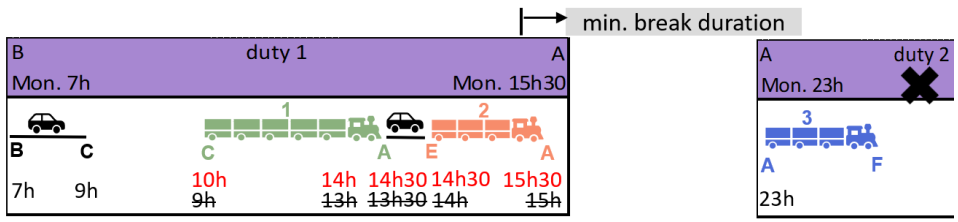
Uncertainties at SNCF: delay propagation and disruptions

Train delays of two kinds:

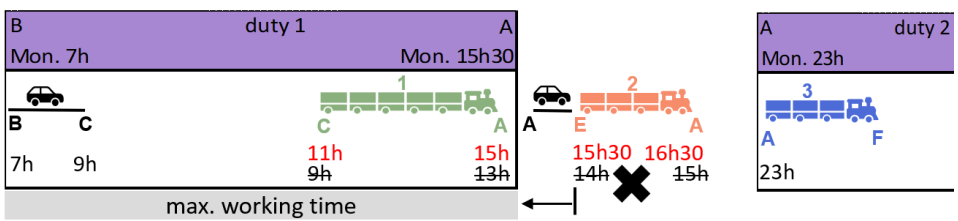
- Primary delays: exogenous (e.g., shunting yard incident, engine failure, etc.)
- Secondary delays: endogenous, from delay propagation within duties

When delays exceed some threshold:

- a break might not meet the minimal duration. In this case, the next duty is disrupted;



- the working time within the duty might be too long. In this case, one or multiple trains are disrupted.



Uncertainties at SNCF: fragility score

Assumption:

- primary delays distributions are known.

H: random variable for arrival times (with primary and secondary delays)
$\sigma(h, b)$: number of disruptions in the realization $H = h$

Block "fragility" score:

$$\text{fragility}(b) := \mathbb{E}\Big[\sigma(H, b)\Big]$$

Roster "fragility" score:

$$\text{fragility}(r) := \sum_{b \in r} \text{fragility}(b)$$

Stochastic crew management: problem statement and model

Stochastic crew management problem:
Input. Trains to be driven every week, teams, threshold S
Output. Covering of trains by blocks with minimum number of working days, each block assigned to a team, such that the overall fragility is below the threshold S

Stochastic crew management: problem statement and model

Stochastic crew management problem:
Input. Trains to be driven every week, teams, threshold S
Output. Covering of trains by blocks with minimum number of working days, each block assigned to a team, such that the overall fragility is below the threshold S

$$\begin{aligned}
\text{minimize} \quad & \sum_{i \in I} \sum_{b \in B_i} w_b x_{b,i} \\
\text{s.t.} \quad & \sum_{i \in I} \sum_{b \in B_i : t \in b} x_{b,i} \geqslant 1 && \forall t \in T \\
& \sum_{i \in I} \sum_{b \in B_i} \mathbb{E}[\sigma(H,b)] x_{b,i} \leqslant S \\
& x_{b,i} \in \{0,1\} && \forall i \in I, \forall b \in B_i
\end{aligned}$$

- $x_{b,i} \in \{0,1\}$: block $b \in B_i$ is selected for team $i \in I$

Column generation and RCSPP

Assumptions:

- primary delays are independent variables (not necessarily identically distributed).
- their supports are finite.

Marginal modifications of the column generation-based algorithm:

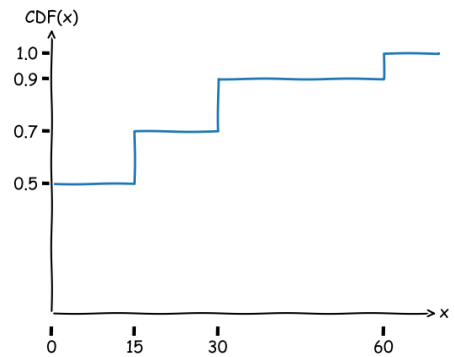
- Farkas lemma based routine to tackle infeasiblilities
- RCSPP bounds based on the usual stochastic order $\leqslant_{\mathrm{st}}$
- Handling delay propagation along a path via distributions convolutions

Numerical experiments: stochastic case

- Run with same setup as the deterministic case
- Column generation time limit 24 h: reached for every instance (stochastic case)
- Integer linear program time limit 1 h

- Threshold S: 50% of two-stage solution

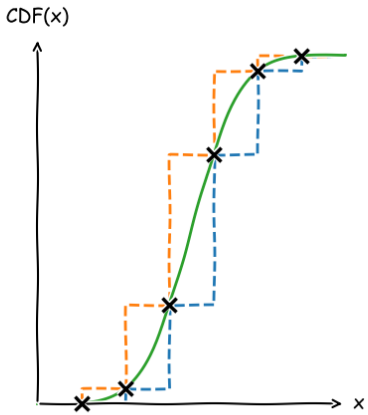
Primary delay cumulative distribution function:



	Two-stage approach		Integrated approach: deterministic case	
	Best found solution	Fragility	Best found solution	Fragility
116 trains	50	0.970	41	2.748
280 trains	228	2.450	207	6.440
926 trains	832	8.538	724	28.460
1,815 trains	1,633	14.900	1,477	58.980

	Integrated approach: stochastic case			
	Best found solution			Fragility
116 trains	(-12%)	44	(-55%)	0.437
280 trains	(-8%)	210	(-51%)	1.202
926 trains	(-11%)	741	(-52%)	4.081
1,815 trains	(-10%)	1,477	(-50%)	7.432

From finite support delay distributions to any distribution

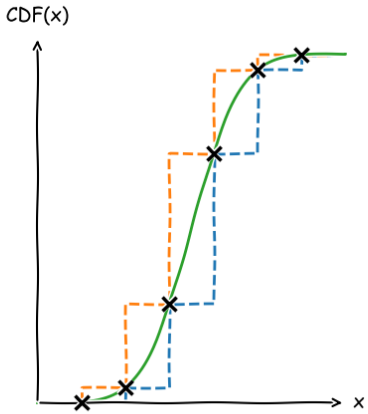


$P'_t \leqslant_{\text{st}} P_t \leqslant_{\text{st}} P''_t$ for every $t \in T$

optimal value under $(P'_t)_t \leqslant$ optimal value under $(P_t)_t$

optimal value under $(P_t)_t \leqslant$ optimal value under $(P''_t)_t$

From finite support delay distributions to any distribution



$P'_t \leqslant_{\text{st}} P_t \leqslant_{\text{st}} P''_t$ for every $t \in T$

optimal value under $(P'_t)_t \leqslant$ optimal value under $(P_t)_t$

optimal value under $(P_t)_t \leqslant$ optimal value under $(P''_t)_t$

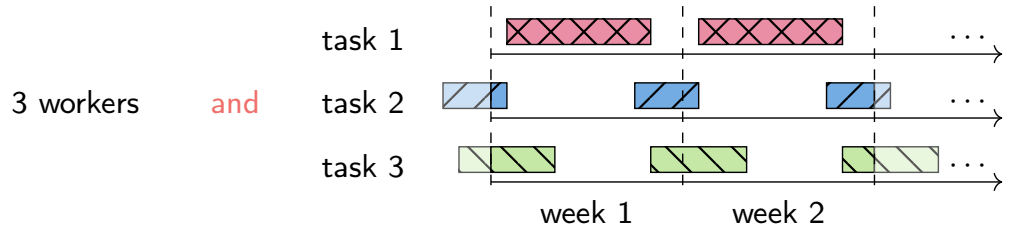
- Original primary delay distributions $(\bar{P}_t)_t$
- Sequence of alternative primary delay distributions $(P_{t,n})_{t,n}$ such that
  - $P_{t,n} \leqslant_{\text{st}} \bar{P}_t$ for every t and n
  - $P_{t,n} \xrightarrow{\text{d}} \bar{P}_t$ for every t

Theorem 5.6.2, p.75

Suppose that the problem is feasible under $(\bar{P}_t)_t$. Then there exists N such that, for all $n \geqslant N$, the sets of feasible solutions under $(P_{t,n})_t$ and under $(\bar{P}_t)_t$ coincide (and the same holds for the set of optimal solutions).

Motivation

- Tasks $[n]$ to be performed at the same time every week
- Number q of indistinguishable workers

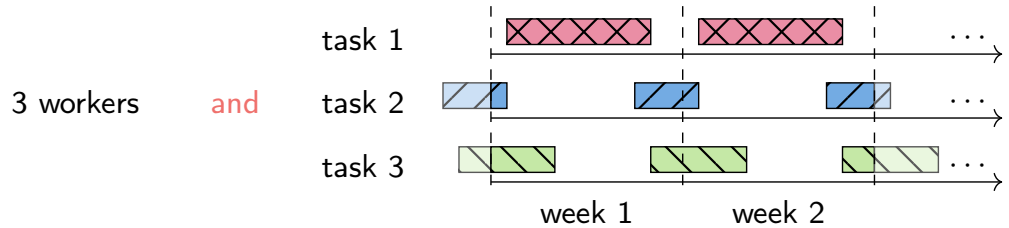


Aim. Find a feasible assignment of tasks to workers so that each worker performs each task with same asymptotic frequency

→ Balanced assignment

Motivation

- Tasks $[n]$ to be performed at the same time every week
- Number q of indistinguishable workers



Aim. Find a feasible assignment of tasks to workers so that each worker performs each task with same asymptotic frequency
→ Balanced assignment

Assignment: $f : [n] \times \mathbb{Z}_{>0} \to [q]$, where $f(i, r) = j$ means worker j performs task i on week r

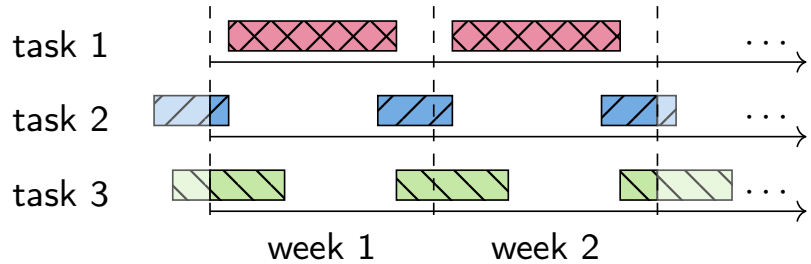
Feasible assignment: f is non overlapping.

Balanced assignment:

$$f \text{ feasible} \quad \text{and} \quad \lim_{t \to +\infty} \frac{1}{t} \left| \{ r \in [t] : f(i, r) = j \} \right| = \frac{1}{q} \qquad \forall i, j$$

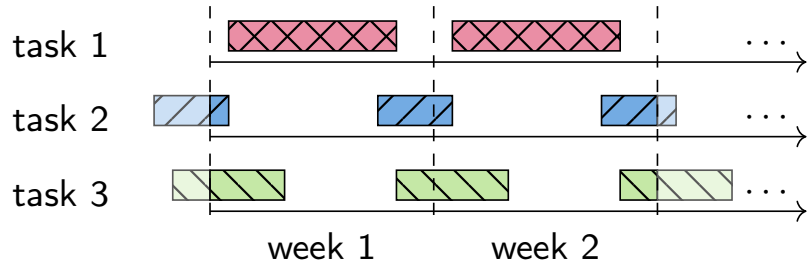
Feasible and balanced assignments: examples

3 workers and

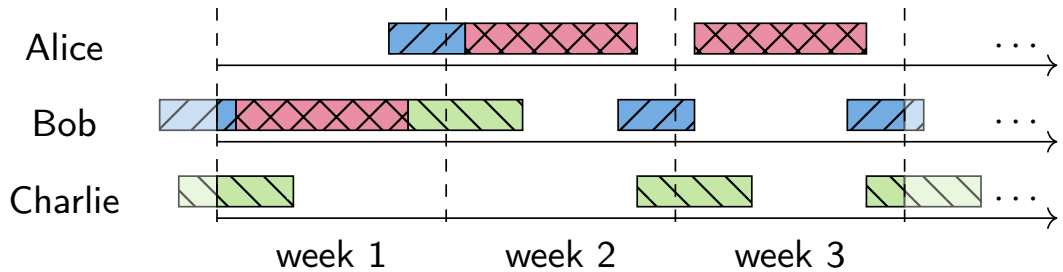


Feasible and balanced assignments: examples

3 workers and

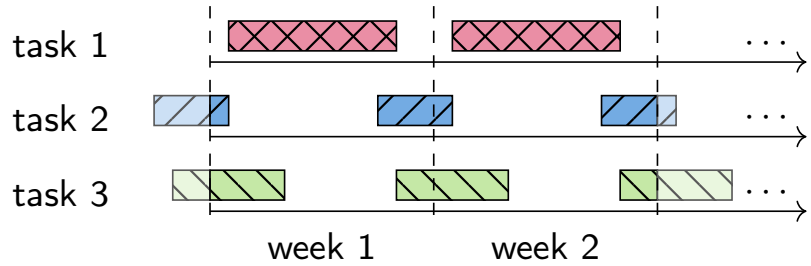


Feasible assignment:

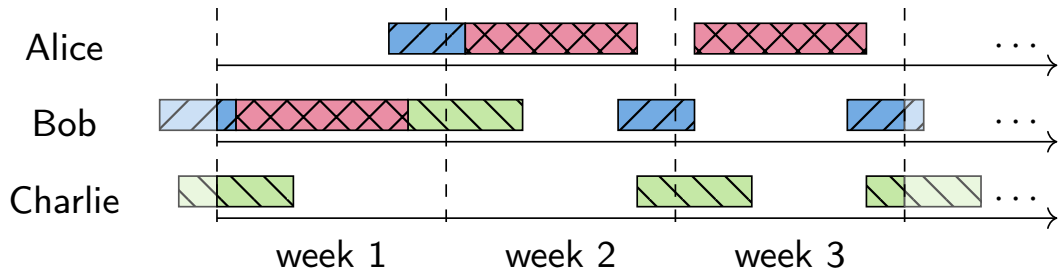


Feasible and balanced assignments: examples

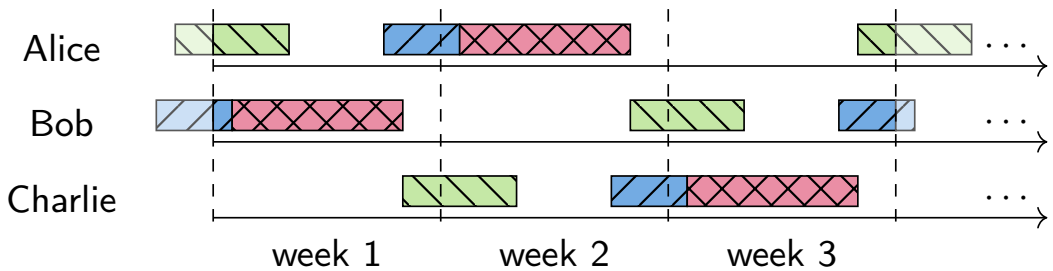
3 workers and



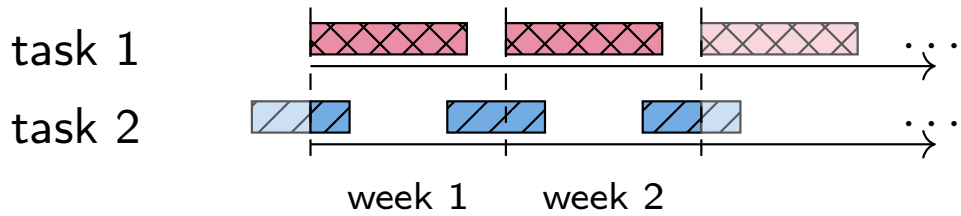
Feasible assignment:



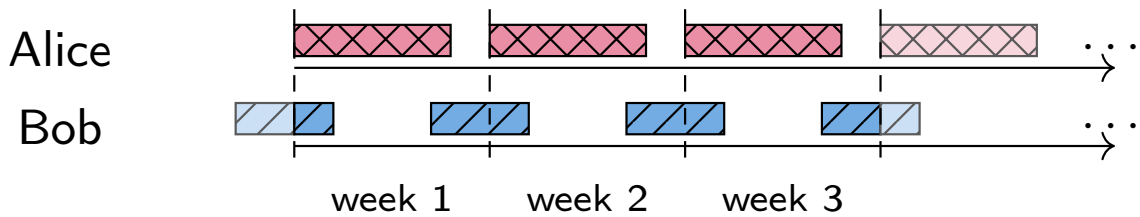
Balanced assignment:



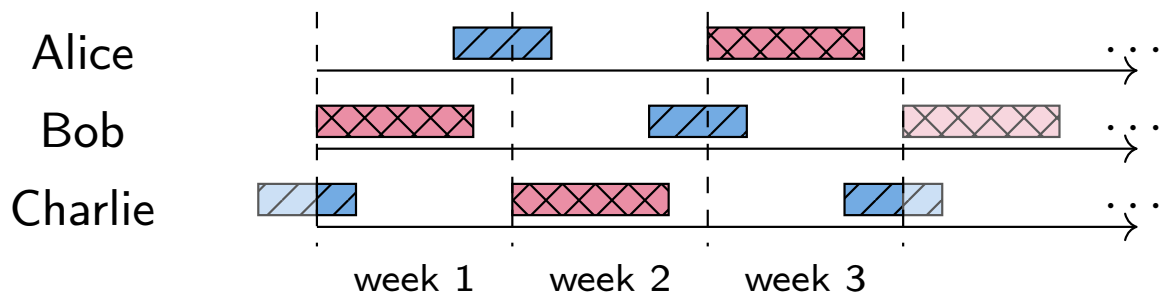
Balanced assignment: unachievable case



With 2 workers

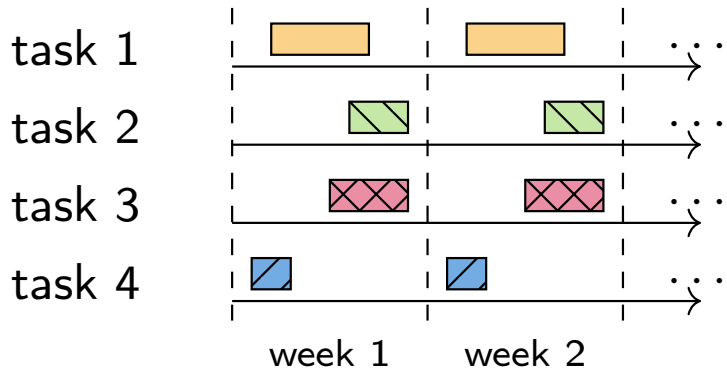


With 3 workers

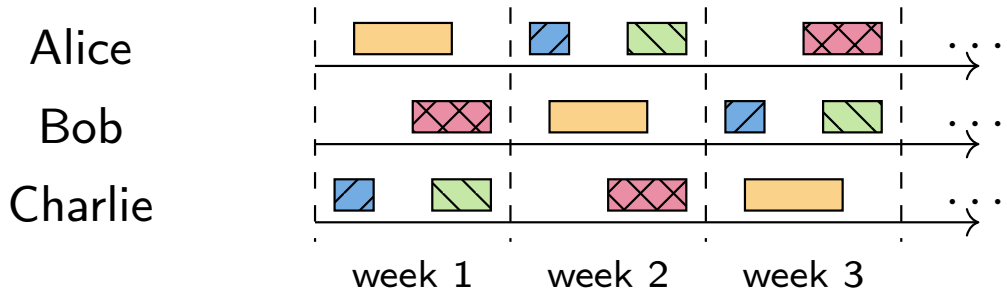


Balanced assignment: trivial case

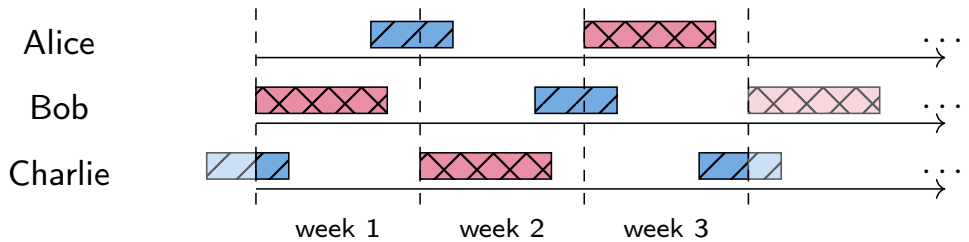
3 workers and



Balanced assignment:



Main results



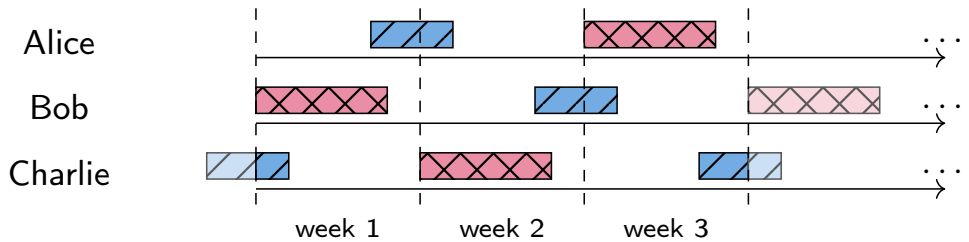
$$\lim_{t \to +\infty} \frac{1}{t} \left| \{ r \in [t] : f(i,r) = j \} \right| = \frac{1}{q} \qquad \forall i, j$$

There exists a balanced assignment.

$\Downarrow$

There exists a feasible assignment with a worker performing each task at least once.

Main results



$$\lim_{t \to +\infty} \frac{1}{t} \left| \{ r \in [t] : f(i,r) = j \} \right| = \frac{1}{q} \qquad \forall i, j$$

There exists a balanced assignment.

$\Updownarrow$

There exists a feasible assignment with a worker performing each task at least once.

Main results

Theorem 7.1.1, p.106

The following statements are equivalent:

1. There exists a feasible assignment with a worker performing each task at least once.
2. There exists a balanced assignment.
3. There exists a periodic balanced assignment with period q.

q: number of workers

Algorithmic proof with polynomiality results:

- Deciding existence of a balanced assignment
- Computing a periodic balanced assignment with period q

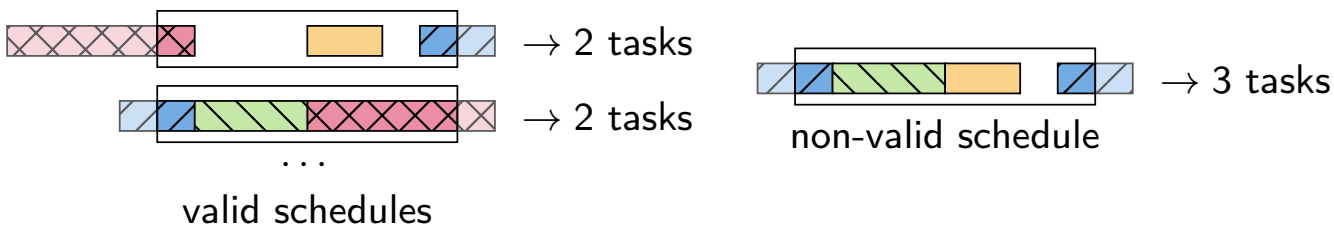
Extended version: setting

Input. n tasks to be repeated every week, q workers, set $\mathcal{S}$ of valid weekly schedules

Feasible assignment: f non overlapping and induces schedules in $\mathcal{S}$

Examples:

- workers perform at most M tasks per week.
  with $M = 2$



- workers work at most H hours per week.
- when q tasks have to be performed at time zero, workers cannot teleport.
- ...

Balanced assignments: results

Theorem 8.1.2, p.126

If there exists a feasible assignment with a worker performing each task at least once and q tasks have to be performed at time zero, then there exists a balanced assignment.

q: number of workers

Examples where the additionnal condition is needed (Fig. 8.1, p.125)

Theorem 8.1.1, p.124

If there exists a balanced assignment, then there exists a periodic balanced assignment with period bounded by $2(q-1)q!$.

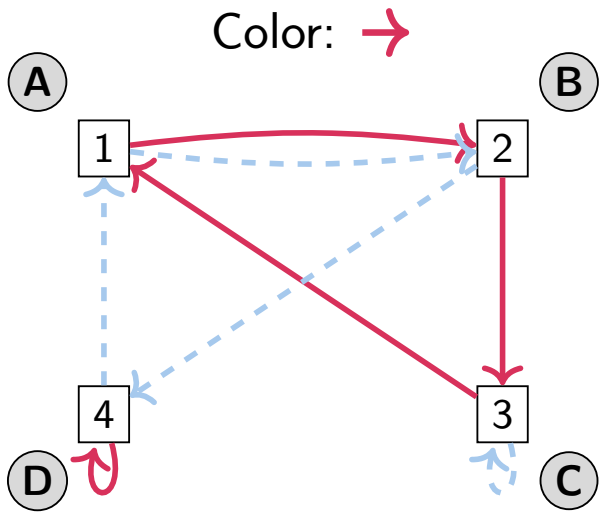
- Existence of a balanced assignment decidable

A problem of pebbles on a graph

Consider:

- graph $D = (V, A)$ with colored arcs, each color forming vertex disjoint cycles covering V
- one pebble on each vertex

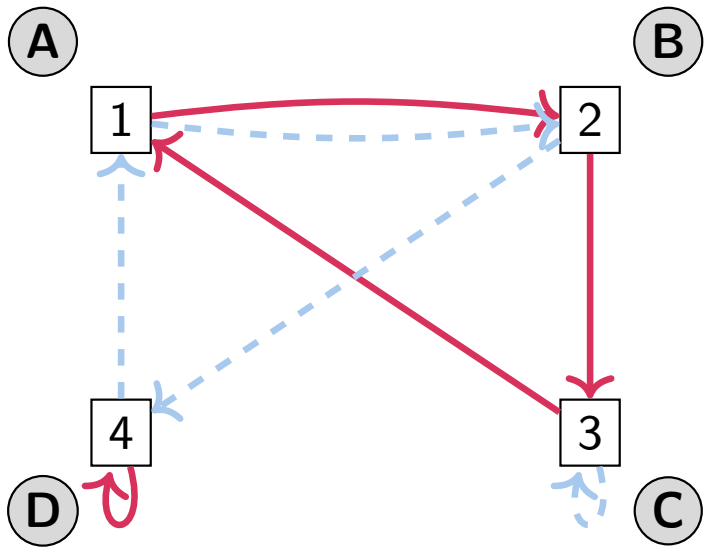
Sequence of colors defines sequence of moves of the pebbles



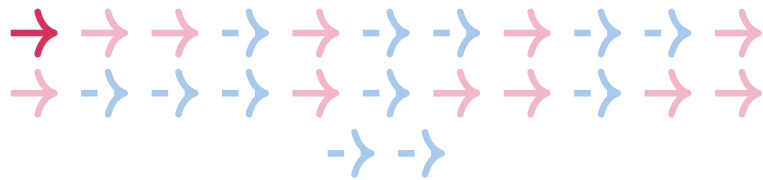
- Vertex = task at time zero
- Arc = valid schedule
- Pebble = worker
- Color c = partition of the tasks into schedules

Balanced sequence of colors: example

Aim. Find a sequence of colors making each pebble visit each arc with same asymptotic frequency.



Sequence of colors:



Arcs visiting:

	1→2	2→3	3→1	4→4	1⇢2	2⇢4	3⇢3	4⇢1
A	**1**	0	0	0	0	0	0	0
B	0	**1**	0	0	0	0	0	0
C	0	0	**1**	0	0	0	0	0
D	0	0	0	**1**	0	0	0	0

Balanced sequence of colors: result

Proposition 8.3.1, p.129

If D is connected, then there exists a sequence of colors making each pebble visit each arc with same asymptotic frequency.

Two proofs:

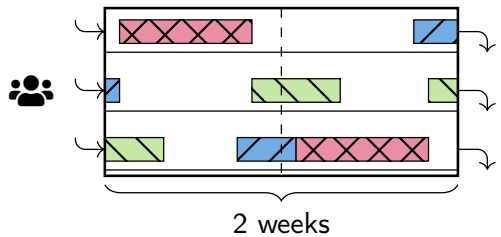
- Markov chains: non constructive
- Eulerian cycle in the Cayley graph associated to the set Σ of colors (seen as permutations of V):
  - Contructive proof
  - Additionnal result: the sequence of colors can be made periodic (period upper bounded by $|A| \times |\langle\Sigma\rangle|/|V|$)

Balanced assignments: graph connected with $2(q-1)$ colors

$\rightarrow$ period upper bounded by $2(q-1)q!$

q: number of workers

What about cyclic rosters?



Corollary 8.5.5, p.135

Suppose q is a prime and there exists a balanced assignment. There exists then a cyclic roster on a k-week basis, with $k \leqslant q! - 1$.

q: number of workers

Using our graph framework along with:

- Group theory tools: Cauchy theorem, orbit-stabilizer theorem

Overall thesis contributions

- Integrating crew scheduling and crew rostering with train delays for rail freight
  - (CHAPTER 4) Integrated crew management
  - (CHAPTER 5) Integrated stochastic crew management
- Balanced assignments of periodic tasks
  - (CHAPTER 7) Basic version: non-overlapping balanced assignments
  - (CHAPTER 8) Extended version: balanced assignments with valid weekly schedules

Overall thesis contributions

- Integrating crew scheduling and crew rostering with train delays for rail freight
  - (CHAPTER 3) Crew scheduling and crew rostering at SNCF
  - (CHAPTER 4) Integrated crew management
  - (CHAPTER 5) Integrated stochastic crew management
- Branch-and-price(-and-cut) for line planning with minimal traveling time
  - (CHAPTER 6) Branch-and-price
- Balanced assignments of periodic tasks
  - (CHAPTER 7) Basic version: non-overlapping balanced assignments
  - (CHAPTER 8) Extended version: balanced assignments with valid weekly schedules
  - (CHAPTER 9) Complementary discussions

Perspectives and open questions

Crew management	Balanced assignments
Multi-objective optimization for the trade-off between fragility and number of working days	Asymptotic bounds from group theory for cyclic rosters
Can we establish an absolute bound on the rank N from which optimal solutions coincide for original and approximate distribution functions?	Is the bound $2(q-1)q!$ for periodic balanced assignments tight for a single instance?

Thank you!